CHAPTER TWO

Linear Regression

Synopsis

Linear regression is introduced as a parameter estimation problem, and least squares solutions are derived. Maximum likelihood is defined, and its association with least squares solutions under normally distributed data errors is demonstrated. Statistical tests based on χ^2 that provide insight into least squares solutions are discussed. The mapping of data errors into model errors in the context of least squares is described. The determination of confidence intervals using the model covariance matrix and the meaning of model parameter correlations is discussed. The problems of estimating unknown data standard deviations and recognizing proportional data errors are addressed. The issue of data outliers and the concept of robust estimation are introduced, and 1-norm minimization is introduced as a robust estimation technique. General propagation of errors between data and model using Monte Carlo methods is discussed in the context of the iteratively reweighted least squares 1-norm minimization algorithm.

2.1. INTRODUCTION TO LINEAR REGRESSION

The problem of finding a parameterized curve that approximately fits a set of data is referred to as **regression**. When the regression model is linear in the fitted parameters, then we have a **linear regression** problem. In this chapter, linear regression problems are analyzed as discrete linear inverse problems.

Consider a discrete linear inverse problem. We begin with a data vector, $\mathbf{d}$, of m observations, and a vector, $\mathbf{m}$, of n model parameters that we wish to determine. The forward problem can be expressed as a linear system of equations using an m by n matrix $\mathbf{G}$

$$\mathbf{Gm} = \mathbf{d}. \tag{2.1}$$

Recall that if $\text{rank}(\mathbf{G}) = n$, then the matrix has full column rank. In this chapter we will assume that the matrix $\mathbf{G}$ has full column rank. In Chapter 3 we will consider rank deficient problems.

For a full column rank matrix, it is frequently the case that there is no solution $\mathbf{m}$ that satisfies (2.1) exactly. This happens because the dimension of the range of $\mathbf{G}$ is smaller than m and a noisy data vector, $\mathbf{d}$, will generally lie outside of the range of $\mathbf{G}$ ($\mathbf{d}$ will lie in R^n for typical noise scenarios).

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A useful approximate solution may still be found by finding a particular model $\mathbf{m}$ that minimizes some measure of the misfit between the actual data and $\mathbf{Gm}$. The **residual vector** is the vector of differences between observed data and corresponding model predictions

$$\mathbf{r} = \mathbf{d} - \mathbf{Gm}. \tag{2.2}$$

and the elements of $\mathbf{r}$ are frequently referred to simply as residuals. One commonly used measure of the misfit is the 2-norm of the residual vector, and a model that minimizes this 2-norm is called a **least squares solution**. The least squares or 2-norm solution is of special interest both because it is readily amenable to analysis and geometric intuition, and because it turns out to be statistically the most likely solution if data errors are normally distributed.

The least squares solution is, from the normal equations (A.73),

$$\mathbf{m}_{L_2} = (\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T\mathbf{d}. \tag{2.3}$$

It can be shown that if $\mathbf{G}$ is of full column rank, then $(\mathbf{G}^T\mathbf{G})^{-1}$ exists (Exercise A.13f).

A classic linear regression problem is finding parameters m_1 and m_2 for a line, $y_i = m_1 + m_2 x_i$, that best fits a set of $m > 2$ data points. The system of equations in this case is

$$\mathbf{Gm} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ . & . \\ . & . \\ . & . \\ 1 & x_m \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ . \\ . \\ . \\ d_m \end{bmatrix} = \mathbf{d}, \tag{2.4}$$

where the d_i are observations of y at each corresponding position x_i. Applying (2.3) to the $\mathbf{G}$ and $\mathbf{m}$ specified in (2.4) gives the least squares solution

$$\begin{aligned} \mathbf{m}_{L_2} &= (\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T\mathbf{d} \\ &= \left(\begin{bmatrix} 1 & \dots & 1 \\ x_1 & \dots & x_m \end{bmatrix} \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & \dots & 1 \\ x_1 & \dots & x_m \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_m \end{bmatrix} \\ &= \begin{bmatrix} m & \sum_{i=1}^m x_i \\ \sum_{i=1}^m x_i & \sum_{i=1}^m x_i^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{i=1}^m d_i \\ \sum_{i=1}^m x_i d_i \end{bmatrix} \\ &= \frac{1}{m \sum_{i=1}^m x_i^2 - (\sum_{i=1}^m x_i)^2} \begin{bmatrix} \sum_{i=1}^m x_i^2 & -\sum_{i=1}^m x_i \\ -\sum_{i=1}^m x_i & m \end{bmatrix} \begin{bmatrix} \sum_{i=1}^m d_i \\ \sum_{i=1}^m x_i d_i \end{bmatrix}. \end{aligned} \tag{2.5}$$

2.2. STATISTICAL ASPECTS OF LEAST SQUARES

If we consider data points to be imperfect measurements that include random errors, then we are faced with the problem of finding the solution that is best from a statistical point of view. One approach, **maximum likelihood estimation**, considers the question from the following perspective. Given that we observed a particular data set, that we know the statistical characteristics of these observations, and that we have a mathematical model for the forward problem, what is the model from which these observations would most likely arise?

Maximum likelihood estimation is a general method that can be applied to any estimation problem where a joint probability density function (B.26) can be assigned to the observations. The essential problem is to find the most likely model, as characterized by the elements of the parameter vector $\mathbf{m}$, for the set of observations contained in the vector $\mathbf{d}$. We will assume that the observations are independent so that we can use the product form of the joint probability density function (B.28).

Given a model $\mathbf{m}$, we have a probability density function $f_i(d_i|\mathbf{m})$ for each of the observations. In general, these probability density functions will vary depending on $\mathbf{m}$, so probability densities are conditional on $\mathbf{m}$. The joint probability density for a vector of independent observations $\mathbf{d}$ will be

$$f(\mathbf{d}|\mathbf{m}) = f_1(d_1|\mathbf{m}) \cdot f_2(d_2|\mathbf{m}) \cdot \cdot \cdot f_m(d_m|\mathbf{m}). \tag{2.6}$$

Note that the $f(d_i|\mathbf{m})$ are probability densities, not probabilities. We can only compute the probability of observing data in some range for a given model $\mathbf{m}$ by integrating $f(\mathbf{d}|\mathbf{m})$ over that range. In fact, the probability of getting any particular set of data exactly is precisely zero! This conceptual conundrum can be resolved by considering the probability of getting a data set that lies within a small m-dimensional box around a particular data set $\mathbf{d}$. This probability will be nearly proportional to the probability density $f(\mathbf{d}|\mathbf{m})$.

In practice, we measure a particular data vector and wish to find the "best" model to match it in the maximum likelihood sense. That is, $\mathbf{d}$ will be a fixed set of observations, and $\mathbf{m}$ will be a vector of parameters to be estimated. The **likelihood function**, L, is the probability of $\mathbf{m}$ given an observed $\mathbf{d}$, which is identical to (2.6), the joint probability density function of $\mathbf{d}$ given $\mathbf{m}$

$$L(\mathbf{m}|\mathbf{d}) = f(\mathbf{d}|\mathbf{m}). \tag{2.7}$$

For many possible models, $\mathbf{m}$, (2.7) will be extremely close to zero because such models would be extremely unlikely to produce the observed data set $\mathbf{d}$. The likelihood would be much larger for any models that, conversely, would be relatively likely to produce the observed data. According to the **maximum likelihood principle** we should select the model $\mathbf{m}$ that maximizes the likelihood function (2.7). Model estimates obtained in this manner have many desirable statistical properties [29, 40].

It is particularly insightful that when we have a discrete linear inverse problem and the data errors are independent and normally distributed, then the maximum likelihood principle solution is the least squares solution. To show this, assume that the data have independent random errors that are normally distributed with expected value zero, and where the standard deviation of the ith observation, d_i, is σ_i. The probability density for d_i then takes the form of (B.6),

$$f_i(d_i|\mathbf{m}) = \frac{1}{\sigma_i\sqrt{2\pi}} e^{-\frac{1}{2}(d_i-(\mathbf{Gm})_i)^2/\sigma_i^2}. \tag{2.8}$$

The likelihood function for the complete data set is the product of the individual likelihoods,

$$L(\mathbf{m}|\mathbf{d}) = \frac{1}{(2\pi)^{m/2}\prod_{i=1}^{m}\sigma_i}\prod_{i=1}^{m} e^{-\frac{1}{2}(d_i-(\mathbf{Gm})_i)^2/\sigma_i^2}. \tag{2.9}$$

The constant factor does not affect the maximization of L, so we can solve

$$\max \prod_{i=1}^{m} e^{-\frac{1}{2}(d_i-(\mathbf{Gm})_i)^2/\sigma_i^2}. \tag{2.10}$$

The natural logarithm is a monotonically increasing function, so we can equivalently solve

$$\max \log \prod_{i=1}^{m} e^{-\frac{1}{2}(d_i-(\mathbf{Gm})_i)^2/\sigma_i^2} = \max \left[-\frac{1}{2}\sum_{i=1}^{m}\frac{(d_i-(\mathbf{Gm})_i)^2}{\sigma_i^2}\right]. \tag{2.11}$$

Finally, if we turn the maximization into a minimization by changing sign and ignore the constant factor of 1/2, the problem becomes

$$\min \sum_{i=1}^{m}\frac{(d_i-(\mathbf{Gm})_i)^2}{\sigma_i^2}. \tag{2.12}$$

Aside from the distinct $1/\sigma_i^2$ factors in each term of the sum, this is identical to the least squares problem for $\mathbf{Gm} = \mathbf{d}$.

To incorporate the data standard deviations into this solution, we scale the system of equations to obtain a weighted system of equations. Let a diagonal weighting matrix be

$$\mathbf{W} = \mathrm{diag}(1/\sigma_1,\ 1/\sigma_2,\ \ldots,\ 1/\sigma_m). \tag{2.13}$$

Then let

$$\mathbf{G}_w = \mathbf{W}\mathbf{G} \tag{2.14}$$

and

$$\mathbf{d}_w = \mathbf{W}\mathbf{d}. \tag{2.15}$$

The weighted system of equations is then

$$\mathbf{G}_w \mathbf{m} = \mathbf{d}_w. \tag{2.16}$$

The normal equations (A.73) solution to (2.16) is

$$\mathbf{m}_{L_2} = (\mathbf{G}_w^T \mathbf{G}_w)^{-1} \mathbf{G}_w^T \mathbf{d}_w. \tag{2.17}$$

Now,

$$\|\mathbf{d}_w - \mathbf{G}_w \mathbf{m}_{L_2}\|_2^2 = \sum_{i=1}^{m} (d_i - (\mathbf{G}\mathbf{m}_{L_2})_i)^2 / \sigma_i^2. \tag{2.18}$$

Thus, the least squares solution to $\mathbf{G}_w \mathbf{m} = \mathbf{d}_w$ is the maximum likelihood solution.

The sum of the squares of the residuals also provides useful statistical information about the quality of model estimates obtained with least squares. The **chi-square statistic** is

$$\chi^2_{\text{obs}} = \sum_{i=1}^{m} (d_i - (\mathbf{G}\mathbf{m}_{L_2})_i)^2 / \sigma_i^2. \tag{2.19}$$

Since χ^2_{obs} depends on the random measurement errors in $\mathbf{d}$, it is itself a random variable. It can be shown that under our assumptions, χ^2_{obs} has a χ^2 distribution with $\nu = m - n$ degrees of freedom [29, 40].

The probability density function for the χ^2 distribution is

$$f_{\chi^2}(x) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\frac{1}{2}\nu - 1} e^{-x/2} \tag{2.20}$$

(Figure B.4). The χ^2 **test** provides a statistical assessment of the assumptions that we used in finding the least squares solution. In this test, we compute χ^2_{obs} and compare it to the theoretical χ^2 distribution with $\nu = m - n$ degrees of freedom.

The probability of obtaining a χ^2 value as large or larger than the observed value (and hence a worse misfit between data and model data predictions than that obtained) is called the ***p*-value** of the test, and is given by

$$p = \int_{\chi^2_{\text{obs}}}^{\infty} f_{\chi^2}(x)\, dx. \tag{2.21}$$

When data errors are independent and normally distributed, and the mathematical model is correct, it can be shown that the p-value will be uniformly distributed between zero and one (Exercise 2.4). In practice, particular p-values that are very close to either extreme indicate that one or more of these assumptions are incorrect.

There are three general cases.

1. The p-value is not too small and not too large. Our least squares solution produces an acceptable data fit and our statistical assumptions of data errors are consistent. Practically, p does not actually have to be very large to be deemed marginally "acceptable" in many cases (e.g., $p \approx 10^{-2}$), as truly "wrong" models will typically produce extraordinarily small p-values (e.g., 10^{-12}) because of the short-tailed nature of the normal distribution.

   Because the p-value will be uniformly distributed when we have a correct mathematical model and our statistical data assumptions are valid, it is inappropriate to conclude anything based on the differences between p-values in this range. For example, one should not conclude that a p-value of 0.7 is "better" than a p-value of 0.2.

2. The p-value is very small. We are faced with three nonexclusive possibilities, but something is clearly wrong.

   a. The data truly represent an extremely unlikely realization. This is easy to rule out for p-values very close to zero. For example, suppose an experiment produced a data realization where the probability of a worse fit was 10^{-9}. If the model was correct, then we would have to perform on the order of a billion experiments to get a comparably poor fit to the data. It is far more likely that something else is wrong.

   b. The mathematical model $\mathbf{Gm} = \mathbf{d}$ is incorrect. Most often this happens because we have left some important aspect of the physics out of the mathematical model.

   c. The data errors are underestimated or not normally distributed. In particular, we may have underestimated the σ_i.

3. The p-value is very close to one. The fit of the model predictions to the data is almost exact. We should investigate the possibility that we have overestimated the data errors. A more sinister possibility is that a very high p-value is indicative of data fraud, such as might happen if data were cooked up ahead of time to fit a particular model!

A rule of thumb for problems with a large number of degrees of freedom, ν, is that the expected value of χ^2 approaches ν. This arises because, by the central limit theorem (Section B.6), the χ^2 random variable, which is itself a sum of random variables, will become normally distributed as the number of terms in the sum becomes large. The mean of the resulting distribution will approach ν and the standard deviation will approach $(2\nu)^{1/2}$.

In addition to examining χ^2_{obs}, it is important to examine the individual weighted residuals corresponding to a model, $\text{r}_{w,i} = (\mathbf{d} - \mathbf{Gm})_i/\sigma_i = (\mathbf{d}_w - \mathbf{G}_w\mathbf{m})_i$. The elements of $\mathbf{r}_w$ should be roughly normally distributed with standard deviation one and should show no obvious patterns. In some cases where an incorrect model has been fitted to the data, the residuals will reveal the nature of the modeling error. For example, in linear regression to a line, it might be that all of the residuals are negative for small and large values of the independent variable x but positive for intermediate values of x. This would suggest that an additional quadratic term is required in the regression model.

Parameter estimates obtained via linear regression are linear combinations of the data (2.17). If the data errors are normally distributed, then the parameter estimates will also be normally distributed because a linear combination of normally distributed random variables is normally distributed [4, 29]. To derive the mapping between data and model covariances, consider the covariance of a data vector, $\mathbf{d}$, of normally distributed, independent random variables, operated on by a general linear transformation specified by a matrix, $\mathbf{A}$. The appropriate covariance mapping is (B.65)

$$\text{Cov}(\mathbf{Ad}) = \mathbf{A}\text{Cov}(\mathbf{d})\mathbf{A}^T. \tag{2.22}$$

The least squares solution has $\mathbf{A} = (\mathbf{G}_w^T\mathbf{G}_w)^{-1}\mathbf{G}_w^T$. The general covariance matrix of the model parameters for a least squares solution is thus

$$\text{Cov}(\mathbf{m}_{L_2}) = (\mathbf{G}_w^T\mathbf{G}_w)^{-1}\mathbf{G}_w^T\text{Cov}(\mathbf{d}_w)\mathbf{G}_w(\mathbf{G}_w^T\mathbf{G}_w)^{-1}. \tag{2.23}$$

If the weighted data are independent, and thus have an identity covariance matrix, this simplifies to

$$\text{Cov}(\mathbf{m}_{L_2}) = (\mathbf{G}_w^T\mathbf{G}_w)^{-1}\mathbf{G}_w^T\mathbf{I}_m\mathbf{G}_w(\mathbf{G}_w^T\mathbf{G}_w)^{-1} = (\mathbf{G}_w^T\mathbf{G}_w)^{-1}. \tag{2.24}$$

In the case of independent and identically distributed normal data errors, so that the data covariance matrix Cov($\mathbf{d}$) is simply the variance σ^2 times the m by m identity matrix, $\mathbf{I}_m$, (2.24) can be written in terms of the unweighted system matrix as

$$\text{Cov}(\mathbf{m}_{L_2}) = \sigma^2(\mathbf{G}^T\mathbf{G})^{-1}. \tag{2.25}$$

Note that even for a diagonal data covariance matrix, the model covariance matrix is typically not diagonal, and the model parameters are thus correlated. Because elements of least squares models are each constructed from linear combinations of the data vector elements, this statistical dependence between the elements of $\mathbf{m}$ should not be surprising.

The expected value of the least squares solution is

$$E[\mathbf{m}_{L_2}] = (\mathbf{G}_w^T\mathbf{G}_w)^{-1}\mathbf{G}_w^TE[\mathbf{d}_w]. \tag{2.26}$$

Because $E[\mathbf{d}_w] = \mathbf{d}_{\text{true},w}$, and $\mathbf{G}_w\mathbf{m}_{\text{true}} = \mathbf{d}_{\text{true},w}$, we have

$$\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}_{\text{true}} = \mathbf{G}_w^T\mathbf{d}_{\text{true},w}. \tag{2.27}$$

Thus

$$E[\mathbf{m}_{L_2}] = (\mathbf{G}_w^T\mathbf{G}_w)^{-1}\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}_{\text{true}} \tag{2.28}$$

$$= \mathbf{m}_{\text{true}}. \tag{2.29}$$

In statistical terms, the least squares solution is said to be **unbiased**.

We can compute 95% confidence intervals for individual model parameters using the fact that each model parameter m_i has a normal distribution with mean given by the corresponding element of $\mathbf{m}_{\text{true}}$ and variance $\text{Cov}(\mathbf{m}_{L_2})_{i,i}$. The 95% confidence intervals are given by

$$\mathbf{m}_{L_2} \pm 1.96\ \text{diag}(\text{Cov}(\mathbf{m}_{L_2}))^{1/2}, \tag{2.30}$$

where the 1.96 factor arises from

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-1.96\sigma}^{1.96\sigma} e^{-\frac{x^2}{2\sigma^2}}\, dx \approx 0.95. \tag{2.31}$$

Example 2.1

Let us recall Example 1.1 of linear regression of ballistic observations to a quadratic model, where the regression model is

$$y(t) = m_1 + m_2 t - (1/2)m_3 t^2. \tag{2.32}$$

Here y is measured in the upward direction, and the minus sign is applied to the third term because gravitational acceleration is downward. Consider a synthetic data set with $m = 10$ observations and independent normal data errors ($\sigma = 8$ m), generated using

$$\mathbf{m}_{\text{true}} = [10\ \text{m},\ 100\ \text{m/s},\ 9.8\ \text{m/s}^2]^T \tag{2.33}$$

(Table 2.1).

To obtain the least squares solution, we construct the $\mathbf{G}$ matrix. The *ith* row of $\mathbf{G}$ is given by

$$\mathbf{G}_{i,\cdot} = [1,\ t_i,\ -(1/2)t_i^2], \tag{2.34}$$

Table 2.1 Data for the ballistics example.

***t* (s)**	1	2	3	4	5	6	7	8	9	10
***y* (m)**	109.4	187.5	267.5	331.9	386.1	428.4	452.2	498.1	512.3	513.0

so that

$$\mathbf{G} = \begin{bmatrix} 1 & 1 & -0.5 \\ 1 & 2 & -2.0 \\ 1 & 3 & -4.5 \\ 1 & 4 & -8.0 \\ 1 & 5 & -12.5 \\ 1 & 6 & -18.0 \\ 1 & 7 & -24.5 \\ 1 & 8 & -32.0 \\ 1 & 9 & -40.5 \\ 1 & 10 & -50.0 \end{bmatrix}. \tag{2.35}$$

We solve for the parameters using the weighted normal equations, (2.17), to obtain a model estimate,

$$\mathbf{m}_{L_2} = [16.4 \text{ m},\ 97.0 \text{ m/s},\ 9.4 \text{ m/s}^2]^T. \tag{2.36}$$

Figure 2.1 shows the observed data and the fitted curve. The model covariance matrix associated with $\mathbf{m}_{L_2}$ is

$$\mathrm{Cov}(\mathbf{m}_{L_2}) = \begin{bmatrix} 88.53 & -33.60 & -5.33 \\ -33.60 & 15.44 & 2.67 \\ -5.33 & 2.67 & 0.48 \end{bmatrix}. \tag{2.37}$$

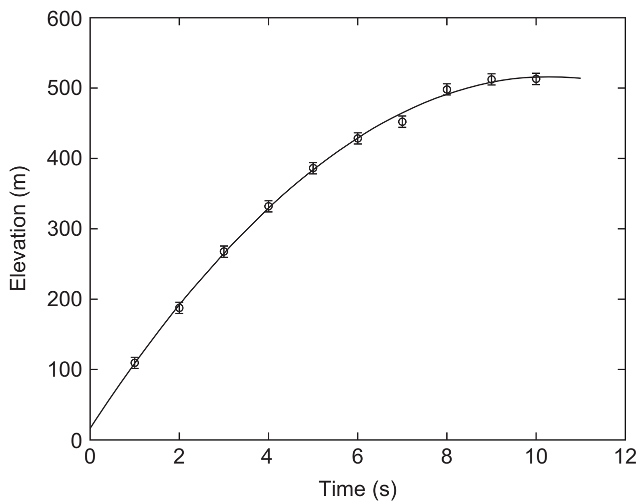

Figure 2.1 Data and model predictions for the ballistics example, with one standard deviation data error bounds indicated.

In our example, $m = 10$ and $n = 3$, so (2.30) gives the following parameter estimates with 95% confidence intervals:

$$\mathbf{m}_{L_2} = [16.42 \pm 18.44\text{m},\ 96.97 \pm 7.70\text{m/s},\ 9.41 \pm 1.36\text{m/s}^2]^T. \tag{2.38}$$

The χ^2 value for this regression is approximately 4.2, and the number of degrees of freedom is $\nu = m - n = 10 - 3 = 7$, so the p-value, (2.21), is

$$p = \int_{4.20}^{\infty} \frac{1}{2^{7/2}\Gamma(7/2)} x^{\frac{5}{2}} e^{-\frac{x}{2}} dx \approx 0.76, \tag{2.39}$$

which is in the realm of plausibility. This means that the fitted model is consistent with the modeling and data uncertainty assumptions.

If we consider combinations of model parameters, the interpretation of the uncertainty in the model parameters becomes more complicated. To characterize model uncertainty more effectively, we can examine 95% **confidence regions** for pairs or larger sets of parameters. When joint parameter confidence regions are projected onto the coordinate axes, m_i, we obtain intervals for parameters that may be significantly larger than we would estimate when considering parameters individually, as in (2.38).

For a vector of estimated model parameters $\mathbf{m}_{L_2}$ characterized by an n-dimensional multivariate normal distribution with mean $\mathbf{m}_{\text{true}}$ and covariance matrix $\mathbf{C} = \text{Cov}(\mathbf{m}_{L_2})$,

$$(\mathbf{m}_{\text{true}} - \mathbf{m}_{L_2})^T \mathbf{C}^{-1} (\mathbf{m}_{\text{true}} - \mathbf{m}_{L_2}), \tag{2.40}$$

can be shown to have a χ^2 distribution with n degrees of freedom [89]. Thus if Δ^2 is the 95th percentile of the χ^2 distribution with n degrees of freedom, the 95% confidence region is defined by the inequality

$$(\mathbf{m}_{\text{true}} - \mathbf{m}_{L_2})^T \mathbf{C}^{-1} (\mathbf{m}_{\text{true}} - \mathbf{m}_{L_2}) \leq \Delta^2. \tag{2.41}$$

The confidence region defined by this inequality is an n-dimensional ellipsoid.

If we wish to find an error ellipsoid for a lower dimensional subset of the model parameters, we can project the n-dimensional error ellipsoid onto the lower dimensional subspace by taking only those rows and columns of $\mathbf{C}$ and elements of $\mathbf{m}$ that correspond to the dimensions that we want to keep [1]. In this case, the number of degrees of freedom in the associated Δ^2 calculation should be correspondingly reduced from n to match the number of model parameters in the projected error ellipsoid.

Since the covariance matrix and its inverse are symmetric and positive definite, we can diagonalize $\mathbf{C}^{-1}$ using (A.77) as

$$\mathbf{C}^{-1} = \mathbf{P}^T \mathbf{\Lambda} \mathbf{P}, \tag{2.42}$$

where $\mathbf{\Lambda}$ is a diagonal matrix of positive eigenvalues and the columns of $\mathbf{P}$ are orthonormal eigenvectors. The semiaxes defined by the columns of $\mathbf{P}$ and associated eigenvalues are referred to as the **principal axes** of the error ellipsoid. The ith semimajor error ellipsoid axis direction is defined by $\mathbf{P}_{\cdot,i}$ and its corresponding length is $\Delta/\sqrt{\Lambda_{i,i}}$.

Because the model covariance matrix is typically not diagonal, the principal axes are typically not aligned in the m_i axis directions. However, we can project the appropriate confidence ellipsoid onto the m_i axes to obtain a "box" that includes the entire 95% error ellipsoid, along with some additional external volume. Such a box provides a conservative confidence interval for a joint collection of model parameters.

Correlations for parameter pairs $(m_i,\ m_j)$ are measures of the inclination of the error ellipsoid with respect to the parameter axes. A correlation approaching $+1$ means the projection is needle-like with its long principal axis having a positive slope, a zero correlation means that the projection has principal axes that are aligned with the axes of the $(m_i,\ m_j)$ plane, and a correlation approaching -1 means that the projection is needle-like with its long principal axis having a negative slope.

Example 2.2

The parameter correlations for Example 2.1 are

$$\rho_{m_i,m_j} = \frac{\mathrm{Cov}(m_i,\ m_j)}{\sqrt{\mathrm{Var}(m_i)\cdot \mathrm{Var}(m_j)}}, \tag{2.43}$$

which give

$$\rho_{m_1,m_2} = -0.91 \tag{2.44}$$

$$\rho_{m_1,m_3} = -0.81 \tag{2.45}$$

$$\rho_{m_2,m_3} = 0.97. \tag{2.46}$$

The three model parameters are highly statistically dependent, and the error ellipsoid is thus inclined in model space. Figure 2.2 shows the 95% confidence ellipsoid.

Diagonalization of $\mathbf{C}^{-1}$ (2.42) shows that the directions of the semiaxes for the error ellipsoid are

$$\mathbf{P} = [\mathbf{P}_{\cdot,1},\ \mathbf{P}_{\cdot,2},\ \mathbf{P}_{\cdot,3}] \approx \begin{bmatrix} 0.93 & 0.36 & -0.03 \\ -0.36 & 0.90 & -0.23 \\ -0.06 & 0.23 & 0.97 \end{bmatrix}, \tag{2.47}$$

with corresponding eigenvalues

$$[\lambda_1,\ \lambda_2,\ \lambda_3] \approx [0.0098,\ 0.4046,\ 104.7]. \tag{2.48}$$

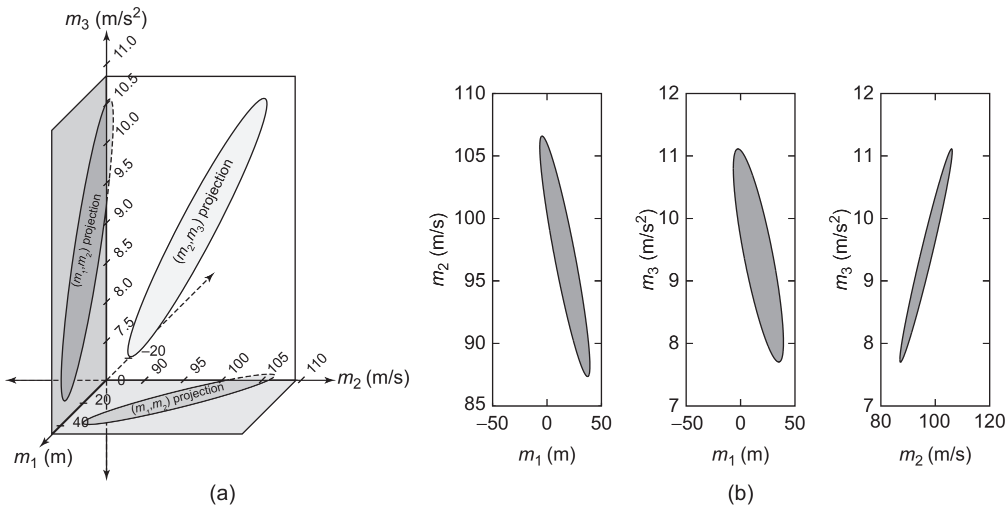

Figure 2.2 Projections of the 95% error ellipsoid onto model axes. (a) Projections in perspective; (b) projections onto the parameter axis planes.

The corresponding 95% confidence ellipsoid semiaxis lengths are

$$\sqrt{F^{-1}_{\chi^2,3}(0.95)}[1/\sqrt{\lambda_1},\ 1/\sqrt{\lambda_2},\ 1/\sqrt{\lambda_3}] \approx [28.2,\ 4.4,\ 0.27] \tag{2.49}$$

where $F^{-1}_{\chi^2,3}(0.95) \approx 7.80$ is the 95th percentile of the χ^2 distribution with three degrees of freedom.

Projecting the 95% confidence ellipsoid defined by (2.47) and (2.49) into the $(m_1,\ m_2,\ m_3)$ coordinate system, and selecting maximum absolute values in the m_i directions to define an ellipsoid-bounding box, we obtain 95% confidence intervals for the parameters considered jointly,

$$[m_1,\ m_2,\ m_3] = [16.42 \pm 26.25\ \text{m},\ 96.97 \pm 10.24\ \text{m/s},\ 9.41 \pm 1.65\ \text{m/s}^2], \tag{2.50}$$

which are appreciably broader than the single parameter confidence estimates obtained using only the diagonal covariance matrix terms in (2.38). Note that there is actually a greater than 95% probability that the box defined by (2.50) will include the true values of the parameters. The reason is that these intervals, considered together as a rectangular prism-shaped region, include a significant parameter space volume that lies outside of the 95% confidence ellipsoid.

It is insightful to note that the model covariance matrix (2.23) does not depend on the estimated model, but depends solely on the system matrix and data covariance. Model covariance is thus exclusively a characteristic of experimental design that reflects how much influence the noise in a *general* data set will have on a model estimate, not on *particular* data values from an individual experiment. It is essential to evaluate the p-value, or another "goodness-of-fit" measure, in assessing a model because an examination of the solution parameters and the covariance matrix alone does *not* reveal whether we are actually fitting the data adequately.

2.3. AN ALTERNATIVE VIEW OF THE 95% CONFIDENCE ELLIPSOID

Recall (2.29) that in linear regression, the least squares solution $\mathbf{m}_{L_2}$ for zero-mean multivariate distributed normal data errors itself has a multivariate normal distribution with

$$E[\mathbf{m}_{L_2}] = \mathbf{m}_{\text{true}}. \tag{2.51}$$

By (2.24), the model covariance matrix is

$$\mathbf{C} = \text{Cov}(\mathbf{m}_{L_2}) = (\mathbf{G}_w^T \mathbf{G}_w)^{-1}, \tag{2.52}$$

where the rows of $\mathbf{G}_w$ are those of $\mathbf{G}$ that have been weighted by respective reciprocal data standard deviations (2.14), and we assume that the data errors are independent and that $(\mathbf{G}_w^T \mathbf{G}_w)$ is nonsingular. By Theorem B.6,

$$(\mathbf{m}_{\text{true}} - \mathbf{m}_{L_2})^T \mathbf{C}^{-1} (\mathbf{m}_{\text{true}} - \mathbf{m}_{L_2}) \tag{2.53}$$

has a χ^2 distribution with degrees of freedom equal to the number of model parameters, n. Let Δ^2 be the 95th percentile of the χ^2 distribution with n degrees of freedom. Then the probability

$$P\left((\mathbf{m}_{\text{true}} - \mathbf{m}_{L_2})^T \mathbf{C}^{-1} (\mathbf{m}_{\text{true}} - \mathbf{m}_{L_2}) \leq \Delta^2\right) \tag{2.54}$$

will be 0.95.

Although (2.54) describes an ellipsoid centered at $\mathbf{m}_{\text{true}}$, the inequality is symmetric in $\mathbf{m}_{\text{true}}$ and $\mathbf{m}_{L_2}$, and can also therefore be thought of as defining an ellipsoid centered around $\mathbf{m}_{L_2}$. Thus, there is a 95% probability that when we gather our data and compute $\mathbf{m}_{L_2}$, the true model $\mathbf{m}_{\text{true}}$ will lie within the model space ellipsoid defined by

$$(\mathbf{m} - \mathbf{m}_{L_2})^T \mathbf{C}^{-1} (\mathbf{m} - \mathbf{m}_{L_2}) \leq \Delta^2. \tag{2.55}$$

Since $\mathbf{C} = (\mathbf{G}_w^T \mathbf{G}_w)^{-1}$, $\mathbf{C}^{-1} = \mathbf{G}_w^T \mathbf{G}_w$, and the 95th percentile confidence ellipsoid can also be written as

$$(\mathbf{m} - \mathbf{m}_{L_2})^T \mathbf{G}_w^T \mathbf{G}_w (\mathbf{m} - \mathbf{m}_{L_2}) \leq \Delta^2. \tag{2.56}$$

If we let

$$\chi^2(\mathbf{m}) = \|\mathbf{G}_w \mathbf{m} - \mathbf{d}_w\|_2^2 = (\mathbf{G}_w \mathbf{m} - \mathbf{d}_w)^T (\mathbf{G}_w \mathbf{m} - \mathbf{d}_w), \tag{2.57}$$

then

$$\begin{aligned}\chi^2(\mathbf{m})-\chi^2(\mathbf{m}_{L_2}) &= (\mathbf{G}_w\mathbf{m}-\mathbf{d}_w)^T(\mathbf{G}_w\mathbf{m}-\mathbf{d}_w)-(\mathbf{G}_w\mathbf{m}_{L_2}-\mathbf{d}_w)^T(\mathbf{G}_w\mathbf{m}_{L_2}-\mathbf{d}_w)\\ &= \mathbf{m}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}-\mathbf{d}_w^T\mathbf{G}_w\mathbf{m}-\mathbf{m}^T\mathbf{G}_w^T\mathbf{d}_w+\mathbf{d}_w^T\mathbf{d}_w\\ &\quad -\mathbf{m}_{L_2}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}_{L_2}+\mathbf{d}_w^T\mathbf{G}_w\mathbf{m}_{L_2}+\mathbf{m}_{L_2}\mathbf{G}_w^T\mathbf{d}_w-\mathbf{d}_w^T\mathbf{d}_w\\ &= \mathbf{m}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}-\mathbf{d}_w^T\mathbf{G}_w\mathbf{m}-\mathbf{m}^T\mathbf{G}_w^T\mathbf{d}_w\\ &\quad -\mathbf{m}_{L_2}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}_{L_2}+\mathbf{d}_w^T\mathbf{G}_w\mathbf{m}_{L_2}+\mathbf{m}_{L_2}^T\mathbf{G}_w^T\mathbf{d}_w. \end{aligned} \tag{2.58}$$

Since $\mathbf{m}_{L_2}$ is a least squares solution to the weighted system of equations, it satisfies the corresponding normal equations. We can thus replace all occurrences of $\mathbf{G}_2^T\mathbf{d}_w$ with $\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}_{L_2}$ using (2.3) to obtain

$$\begin{aligned}\chi^2(\mathbf{m})-\chi^2(\mathbf{m}_{L_2}) &= \mathbf{m}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}-\mathbf{m}_{L_2}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}-\mathbf{m}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}_{L_2}\\ &\quad -\mathbf{m}_{L_2}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}_{L_2}+\mathbf{m}_{L_2}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}_{L_2}+\mathbf{m}_{L_2}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}_{L_2}\\ &= \mathbf{m}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}-\mathbf{m}_{L_2}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}\\ &\quad -\mathbf{m}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}_{L_2}+\mathbf{m}_{L_2}^T\mathbf{G}_w^T\mathbf{G}_w\mathbf{m}_{L_2}, \end{aligned} \tag{2.59}$$

and, finally,

$$\chi^2(\mathbf{m})-\chi^2(\mathbf{m}_{L_2}) = (\mathbf{m}-\mathbf{m}_{L_2})^T\mathbf{G}_w^T\mathbf{G}_w(\mathbf{m}-\mathbf{m}_{L_2}). \tag{2.60}$$

Thus our 95% confidence ellipsoid can also be written as

$$\chi^2(\mathbf{m})-\chi^2(\mathbf{m}_{L_2}) \le \Delta^2 \tag{2.61}$$

and the contour of the $\chi^2(\mathbf{m})$ function at $\chi^2(\mathbf{m}_{L_2})+\Delta^2$ gives the boundary of the 95th percentile confidence ellipsoid.

2.4. UNKNOWN MEASUREMENT STANDARD DEVIATIONS

Suppose that we do not know the standard deviations of the measurement errors a priori. In this case, if we assume that the measurement errors are independent and normally distributed with expected value of zero and standard deviation σ, then we can perform the linear regression and estimate σ from the residuals.

First, we find the least squares solution to the unweighted problem $\mathbf{Gm}=\mathbf{d}$, and let

$$\mathbf{r}=\mathbf{d}-\mathbf{G}\mathbf{m}_{L_2}. \tag{2.62}$$

To estimate the standard deviation from the residuals, let

$$s=\frac{||\mathbf{r}||_2}{\sqrt{\nu}}, \tag{2.63}$$

where $\nu = m - n$ is the number of degrees of freedom [40].

As you might expect, there is a statistical cost associated with not knowing the true standard deviation. If the data standard deviations are known ahead of time, then each

$$m_i' = \frac{m_i - m_{\text{true}_i}}{\sqrt{\mathbf{C}_{i,i}}}, \tag{2.64}$$

where $\mathbf{C}$ is the covariance matrix (2.25), has a standard normal distribution. However, if instead of a known standard deviation we have an estimate, s, obtained using (2.63), then, if $\mathbf{C}'$ is given by the covariance matrix formula (2.25), but with $\sigma = s$, each

$$m_i' = \frac{m_i - m_{\text{true}_i}}{\sqrt{\mathbf{C}'_{i,i}}} \tag{2.65}$$

has a Student's t distribution (B.7) with $\nu = m - n$ degrees of freedom. For smaller numbers of degrees of freedom this produces appreciably broader confidence intervals than the standard normal distribution. As ν becomes large, (2.63) becomes an increasingly better estimate of σ as the two distributions converge. Confidence ellipsoids corresponding to this case can also be computed, but the formula is somewhat more complicated than in the case of known standard deviations [40].

In assessing goodness-of-fit in this case, a problem arises in that we can no longer apply the χ^2 test. Recall that the χ^2 test was based on the assumption that the data errors were normally distributed with known standard deviations σ_i. If the actual residuals were too large relative to the σ_i, then χ^2 would be large, and we would reject the linear regression fit based on a very small p-value. However, if we substitute (2.63) into (2.19), we find that $\chi^2_{\text{obs}} = \nu$, so such a model will always pass the χ^2 test.

Example 2.3

Consider the analysis of a linear regression problem in which the measurement errors are assumed to be independent and normally distributed, with equal but unknown standard deviations, σ. We are given data y_i collected at points x_i (Figure 2.3) that appear to follow a linear relationship.

In this case, the system matrix for the forward problem is

$$\mathbf{G} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ . & . \\ . & . \\ . & . \\ 1 & x_m \end{bmatrix}. \tag{2.66}$$

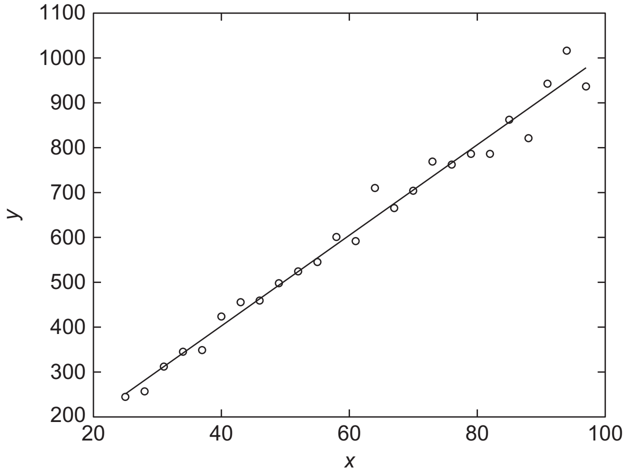

Figure 2.3 Data for Example 2.3, and corresponding linear regression line.

The least squares solution to

$$\mathbf{Gm} = \mathbf{y} \tag{2.67}$$

has

$$y_i = -1.03 + 10.09x_i. \tag{2.68}$$

Figure 2.3 shows the data and the linear regression line. Our estimate of the standard deviation of the measurement errors from (2.63) is $s = 30.74$. The estimated covariance matrix for the fitted parameters is

$$\mathbf{C}' = s^2(\mathbf{G}^T\mathbf{G})^{-1} = \begin{bmatrix} 338.24 & -4.93 \\ -4.93 & 0.08 \end{bmatrix}. \tag{2.69}$$

The parameter confidence intervals, evaluated for each parameter separately, are

$$m_1 = -1.03 \pm \sqrt{338.24}\,t_{m-2,0.975} = -1.03 \pm 38.05 \tag{2.70}$$

and

$$m_2 = 10.09 \pm \sqrt{0.08}\,t_{m-2,0.975} = 10.09 \pm 0.59. \tag{2.71}$$

Since the actual standard deviation of the measurement errors is unknown, we cannot perform a χ^2 test of goodness-of-fit. However, we can still examine the residuals. Figure 2.4 shows the residuals. It is clear that although they appear to be random, the standard deviation seems to increase as x and y increase. This is a common phenomenon

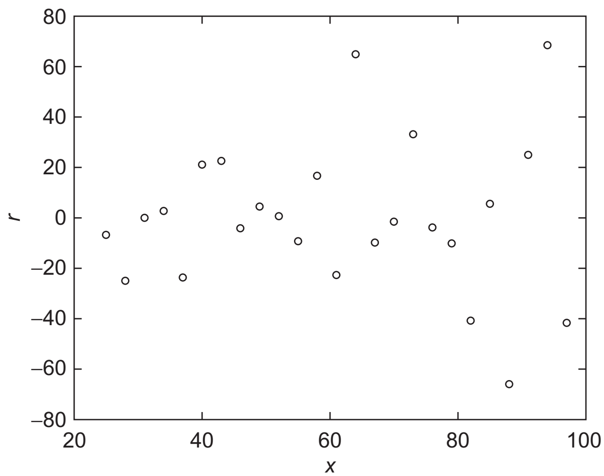

Figure 2.4 Unweighted residuals for Example 2.3.

in linear regression, called a **proportional effect**. One possible way that such an effect might occur is if the sizes of the measurement errors were proportional to the measurement magnitude due to characteristics of the instrument used.

For independent errors where the standard deviations of the data points are proportional to the observation, we can rescale the system of equations (2.67) by dividing each equation by y_i, to obtain

$$\mathbf{G}_w \mathbf{m} = \mathbf{y}_w. \tag{2.72}$$

If statistical assumptions are correct, (2.72) has a least squares solution that approximates (2.17). We obtain a revised least squares estimate of

$$y_i = -12.24 + 10.25 x_i \tag{2.73}$$

with 95% parameter confidence intervals, evaluated as in (2.70) and (2.71), of

$$m_1 = -12.24 \pm 22.39 \tag{2.74}$$

and

$$m_2 = 10.25 \pm 0.47. \tag{2.75}$$

Figure 2.5 shows the data and least squares fit. Figure 2.6 shows the scaled residuals. Note that there is now no obvious trend in the magnitude of the residuals as x and y increase, as there was in Figure 2.4. The estimated standard deviation is 0.045, or 4.5% of the y value. In fact, these data were generated according to the true model

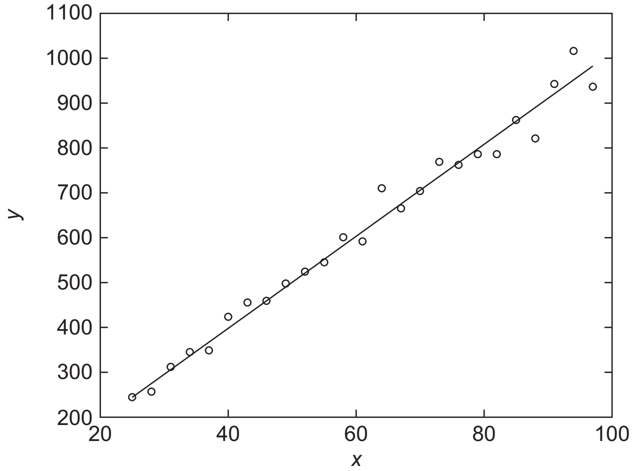

Figure 2.5 Data for Example 2.3, and corresponding linear regression line, weighted system.

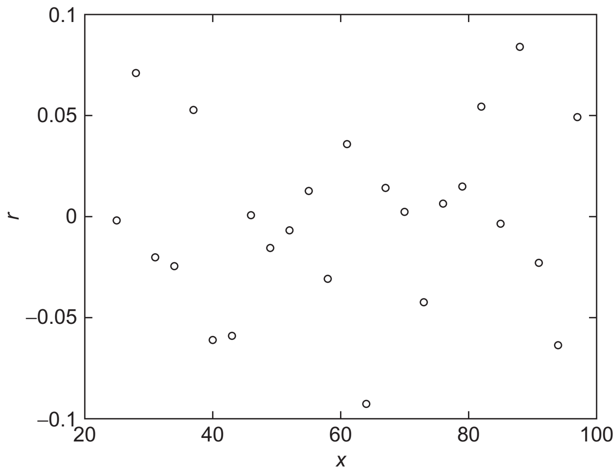

Figure 2.6 Weighted residuals for Example 2.3.

$y_i = 10x_i + 0$, using standard deviations for the measurement errors that were 5% of the y value.

2.5. L_1 REGRESSION

Least squares solutions are highly susceptible to even small numbers of discordant observations, or **outliers**. Outliers are data points that are highly inconsistent with the other data. Outliers may arise from procedural measurement error, for example from incorrectly recording the position of a decimal point in an observation or from instrumental glitches. Outliers should be investigated carefully, since the data may actually be showing us that the form of the mathematical model that we are trying to fit is incorrect. However, if we conclude that there are only a small number of outliers in the data due to incorrect measurements, we need to analyze the data in a way that minimizes their effect on the estimated model.

We can readily appreciate the strong effect of outliers on least squares solutions from a maximum likelihood perspective by noting the very rapid fall-off of the tails of the normal distribution. For example, the probability of a single data point drawn from a normal distribution being more than five standard deviations away from its expected value is less than 1 in 1 million:

$$P(|X - E[X]| \geq 5\sigma) = \frac{2}{\sqrt{2\pi}} \int_5^\infty e^{-\frac{1}{2}x^2}\, dx \approx 6 \times 10^{-7}. \tag{2.76}$$

If an outlier occurs in the data set due to a non–normal error process, the least squares solution will go to great lengths to accommodate it, and thus prevent its corresponding factor in the total likelihood product (2.9) from being vanishingly small.

As an alternative to least squares, consider the solution that minimizes the 1-norm of the residual vector,

$$\mu^{(1)} = \sum_{i=1}^{m} \frac{|d_i - (\mathbf{Gm})_i|}{\sigma_i} = \|\mathbf{d}_w - \mathbf{G}_w\mathbf{m}\|_1. \tag{2.77}$$

The **1-norm solution**, $\mathbf{m}_{L_1}$, will be more outlier resistant, or **robust**, than the least squares solution, $\mathbf{m}_{L_2}$, because (2.77) does not square each of the terms in the misfit measure, as (2.12) does. The 1-norm solution $\mathbf{m}_{L_1}$ also has a maximum likelihood interpretation; it is the maximum likelihood estimator for data with errors distributed according to a double-sided exponential distribution (Appendix B),

$$f(x) = \frac{1}{\sigma\sqrt{2}} e^{-\sqrt{2}|x-\mu|/\sigma}. \tag{2.78}$$

Data sets distributed as (2.78) are unusual. Nevertheless, it is often worthwhile to consider a solution where (2.77) is minimized rather than (2.12), even if most of the measurement errors are normally distributed, should there be reason to suspect the presence of outliers. This solution strategy may be useful if the data outliers occur for reasons that do not undercut our belief that the mathematical model is otherwise correct.

Example 2.4

We can demonstrate the advantages of 1-norm minimization using the quadratic regression example discussed earlier. Figure 2.7 shows the original sequence of independent data points with unit standard deviations, except one of the points (d_4) is now an outlier with respect to a mathematical model of the form (2.32). It is the original data point with 200 m subtracted from it. The least squares model for this data set is

$$\mathbf{m}_{L_2} = [26.4 \text{ m},\ 75.6 \text{ m/s},\ 4.86 \text{ m/s}^2]^T. \tag{2.79}$$

The least squares solution is skewed away from the majority of data points in trying to accommodate the outlier and is a poor estimate of the true model. We can also see that (2.79) fails to fit these data acceptably because of its huge χ^2 value (≈ 489). This is clearly astronomically out of bounds for a problem with 7 degrees of freedom, where the χ^2 value should not be far from 7. The corresponding p-value for $\chi^2 = 489$ is effectively zero.

The upper curve in Figure 2.7

$$\mathbf{m}_{L_1} = [17.6 \text{ m},\ 96.4 \text{ m/s},\ 9.31 \text{ m/s}^2]^T \tag{2.80}$$

is obtained using the 1-norm solution that minimizes (2.77). The data prediction from (2.80) faithfully fits the quadratic trend for the majority of the data points and is only slightly influenced by the outlier at $t = 4$. It is also much closer than (2.79) to the true model (2.33), and to the least squares model for the data set without the outlier (2.36).

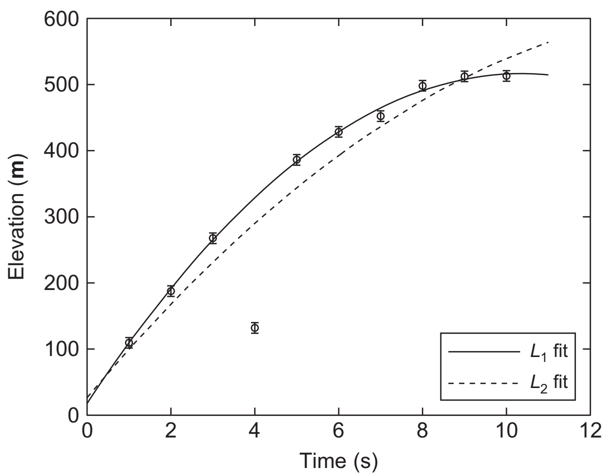

Figure 2.7 L_1 (solid) and L_2 (dashed) solutions for a parabolic data set with an outlier at $t = 4$ s.

In examining the differences between 2- and 1-norm models, it is instructive to consider the almost trivial regression problem of estimating the value of a single parameter from repeated measurements. The system of equations $\mathbf{Gm} = \mathbf{d}$ is

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ . \\ . \\ . \\ 1 \end{bmatrix} \mathbf{m} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ . \\ . \\ . \\ d_m \end{bmatrix}, \tag{2.81}$$

where $\mathbf{m}$ is the 1 by 1 vector containing the parameter that we wish to estimate.

The least squares solution to (2.81) can be seen from the normal equations (A.73) to be simply the observational average

$$\mathbf{m}_{L_2} = (\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T\mathbf{d} = m^{-1}\sum_{i=1}^{m} d_i. \tag{2.82}$$

Finding the 1-norm solution is more complicated. The problem is that the 1-norm of the residual vector,

$$f(\mathbf{m}) = \|\mathbf{d} - \mathbf{Gm}\|_1 = \sum_{i=1}^{m} |d_i - (\mathbf{Gm})_i|, \tag{2.83}$$

is a nondifferentiable function of $\mathbf{m}$ at each point where $(\mathbf{Gm})_i = d_i$. The good news is that $f(\mathbf{m})$ is a convex function of $\mathbf{m}$. Thus any local minimum point is also a global minimum point. We can proceed by finding $f'(\mathbf{m})$ at those points where it is defined, and then separately consider the points at which the derivative is not defined. Every minimum point must either have $f'(\mathbf{m})$ undefined or $f'(\mathbf{m}) = 0$.

At those points where $f'(\mathbf{m})$ is defined, it is given by

$$f'(\mathbf{m}) = \sum_{i=1}^{m} \operatorname{sgn}(d_i - \mathbf{m}). \tag{2.84}$$

where the **signum function**, $\operatorname{sgn}(x)$, is -1 if its argument is negative, 1 if its argument is positive, and 0 if its argument is zero. The derivative (2.84) is zero when exactly half of the data are less than $\mathbf{m}$ and half of the data are greater than $\mathbf{m}$. Of course, this can only happen when the number of observations, m, is even. In this case, any value of $\mathbf{m}$ lying between the two middle observations is a 1-norm solution. When there are an odd number of data, the median data point is the unique 1-norm solution. Even an extreme outlier will not have a large effect on the median of an otherwise clustered set of observations. This illuminates the robustness of the 1-norm solution.

The general problem of finding solutions that minimize $\|\mathbf{d} - \mathbf{Gm}\|_1$ is a bit complicated. One practical method is **iteratively reweighted least squares**, or **IRLS** [138]. The IRLS algorithm solves a sequence of weighted least squares problems whose solutions converge to a 1-norm minimizing solution. Beginning with the residual vector

$$\mathbf{r} = \mathbf{d} - \mathbf{Gm}, \tag{2.85}$$

we want to minimize

$$f(\mathbf{m}) = \|\mathbf{r}\|_1 = \sum_{i=1}^{m} |r_i|. \tag{2.86}$$

The function in (2.86), like the function in (2.83), is nondifferentiable at any point where one of the elements of $\mathbf{r}$ is zero. Ignoring this issue for a moment, we can go ahead and compute the derivatives of f at other points:

$$\frac{\partial f(\mathbf{m})}{\partial m_k} = \sum_{i=1}^{m} \frac{\partial |r_i|}{\partial m_k} = -\sum_{i=1}^{m} G_{i,k}\,\mathrm{sgn}(r_i). \tag{2.87}$$

Writing $\mathrm{sgn}(r_i)$ as $r_i/|r_i|$ gives

$$\frac{\partial f(\mathbf{m})}{\partial m_k} = -\sum_{i=1}^{m} G_{i,k}\frac{1}{|r_i|}r_i. \tag{2.88}$$

The gradient of f is

$$\nabla f(\mathbf{m}) = -\mathbf{G}^T\mathbf{Rr} = -\mathbf{G}^T\mathbf{R}(\mathbf{d} - \mathbf{Gm}), \tag{2.89}$$

where $\mathbf{R}$ is a diagonal weighting matrix with diagonal elements that are the absolute values of the reciprocals of the residuals, so that

$$R_{i,i} = 1/|r_i|. \tag{2.90}$$

To find the 1-norm minimizing solution, we solve $\nabla f(\mathbf{m}) = \mathbf{0}$, which gives

$$\mathbf{G}^T\mathbf{R}(\mathbf{d} - \mathbf{Gm}) = \mathbf{0} \tag{2.91}$$

or

$$\mathbf{G}^T\mathbf{RGm} = \mathbf{G}^T\mathbf{Rd}. \tag{2.92}$$

Because $\mathbf{R}$ is a nonlinear function of $\mathbf{m}$, (2.92) is a nonlinear system of equations that we cannot solve directly. IRLS is an iterative algorithm to find the appropriate weights and solve the system. The algorithm begins with the least squares solution $\mathbf{m}^0 = \mathbf{m}_{L_2}$. We calculate the corresponding residual vector $\mathbf{r}^0 = \mathbf{d} - \mathbf{Gm}^0$ to construct the weighting matrix $\mathbf{R}$ using (2.90). We then solve (2.92) to obtain a new model $\mathbf{m}^1$ and associated residual vector $\mathbf{r}^1$. The process is repeated until the model and residual vectors converge. A typical rule is to stop the iteration when

$$\frac{\|\mathbf{m}^{k+1} - \mathbf{m}^k\|_2}{1 + \|\mathbf{m}^{k+1}\|_2} < \tau \tag{2.93}$$

for some tolerance τ.

If any element of the residual vector becomes zero, then (2.90) becomes undefined. However, this problem can be easily addressed by selecting a tolerance ϵ below which we consider the residuals to be effectively zero. If $|r_i| < \epsilon$, then we set $R_{i,i} = 1/\epsilon$. With this modification it can be shown that this procedure will always converge to an approximate 1-norm minimizing solution.

As with the χ^2 misfit measure for least squares solutions, there is a corresponding p-value that can be evaluated under the assumption of normal data errors for the assessment of 1-norm solutions [127]. Let

$$\mu_{\text{obs}}^{(1)} = \|\mathbf{G}\mathbf{m}_{L_1} - \mathbf{d}\|_1. \tag{2.94}$$

For an observed 1-norm misfit measure (2.94), the probability that a worse misfit could have occurred, given independent and normally distributed data and ν degrees of freedom, is approximately given by

$$p^{(1)}(x, \nu) = P\left(\mu^{(1)} > \mu_{\text{obs}}^{(1)}\right) = 1 - S(x) + \frac{\gamma Z^{(2)}(x)}{6}, \tag{2.95}$$

where

$$S(x) = \frac{1}{\sigma_1\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{\xi^2}{2\sigma_1^2}}\, d\xi \tag{2.96}$$

$$\sigma_1 = \sqrt{(1 - 2/\pi)\nu} \tag{2.97}$$

$$\gamma = \frac{2 - \pi/2}{(\pi/2 - 1)^{3/2}\nu^{\frac{1}{2}}} \tag{2.98}$$

$$Z^{(2)}(x) = \frac{x^2 - 1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \tag{2.99}$$

$$x = \frac{\mu^{(1)} - \sqrt{2/\pi}\,\nu}{\sigma_1}. \tag{2.100}$$

2.6. MONTE CARLO ERROR PROPAGATION

For solution techniques that are nonlinear and/or algorithmic, such as IRLS, there is typically no simple way to propagate uncertainties in the data to uncertainties in the estimated model parameters. In such cases, one can apply **Monte Carlo error propagation** techniques, in which we simulate a collection of noisy data vectors and then examine the statistics of the resulting ensemble of models.

For L_1 minimizing solutions, we can obtain an approximate covariance matrix by first forward-propagating the solution into an assumed noise-free baseline data vector

$$\mathbf{G}\mathbf{m}_{L_1} = \mathbf{d}_b. \tag{2.101}$$

We next re-solve the IRLS problem many times for independent data realizations, obtaining a suite of q 1-norm solutions to

$$\mathbf{G}\mathbf{m}_{L_1,i} = \mathbf{d}_b + \boldsymbol{\eta}_i \tag{2.102}$$

where $\boldsymbol{\eta}_i$ is the ith noise vector realization. If $\mathbf{A}$ is the q by n matrix where the ith row contains the difference between the ith model estimate and the average model,

$$\mathbf{A}_{i,\cdot} = \mathbf{m}_{L_1,i}^T - \bar{\mathbf{m}}_{L_1}^T, \tag{2.103}$$

then an empirical estimate of the covariance matrix is

$$\mathrm{Cov}(\mathbf{m}_{L_1}) = \frac{\mathbf{A}^T\mathbf{A}}{q}. \tag{2.104}$$

Example 2.5

Recall Example 2.4. An estimate of $\mathrm{Cov}(\mathbf{m}_{L_1})$ using 10,000 iterations of the Monte Carlo procedure is

$$\mathrm{Cov}(\mathbf{m}_{L_1}) = \begin{bmatrix} 124.1 & -45.47 & -7.059 \\ -45.47 & 20.54 & 3.501 \\ -7.059 & 3.501 & 0.6316 \end{bmatrix}, \tag{2.105}$$

which contains elements that are about 1.4 times as large as those of the least squares solution (2.37). Unlike least squares solutions, model parameters obtained with the IRLS algorithm will not generally be normally distributed. However, we can compute approximate confidence intervals for the parameters from the covariance matrix diagonal, provided that analysis of the obtained Monte Carlo solution parameter distributions reveals that they are approximately normally distributed. Such an analysis can be performed by examining the parameter distributions with a **Q–Q** plot (Appendix B) and/or generating an ellipsoidal confidence boundary under normal assumptions and counting the proportion of points within the ellipsoid to check for consistency. In this example, **Q–Q** plots reveal the estimates to be approximately normally distributed, and calculating corresponding 95% confidence intervals from (2.105) using (2.30) gives

$$\mathbf{m}_{L_1} = [17.6 \pm 21.8 \text{ m},\ 96.4 \pm 8.88 \text{ m/s},\ 9.31 \pm 1.56 \text{ m/s}^2]^T. \tag{2.106}$$

2.7. EXERCISES

1. A seismic profiling experiment is performed where the first arrival times of seismic energy from a mid-crustal refractor are observed at distances (in kilometers) of

$$\mathbf{x} = \begin{bmatrix} 6.0000 \\ 10.1333 \\ 14.2667 \\ 18.4000 \\ 22.5333 \\ 26.6667 \end{bmatrix} \tag{2.107}$$

from the source, and are found to be (in seconds after the source origin time)

$$\mathbf{t} = \begin{bmatrix} 3.4935 \\ 4.2853 \\ 5.1374 \\ 5.8181 \\ 6.8632 \\ 8.1841 \end{bmatrix}. \tag{2.108}$$

These vectors can also be found in the MATLAB data file **profile.mat**. A two-layer, flat Earth structure gives the mathematical model

$$t_i = t_0 + s_2 x_i, \tag{2.109}$$

where the intercept time, t_0, depends on the thickness and slowness of the upper layer, and s_2 is the slowness of the lower layer. The estimated noise in the first arrival time measurements is believed to be independent and normally distributed with expected value 0 and standard deviation $\sigma = 0.1$ s.

a. Find the least squares solution for the model parameters t_0 and s_2. Plot the data, the fitted model, and the residuals.

b. Calculate and comment on the model parameter correlation matrix (e.g., 2.43). How are the correlations manifested in the general appearance of the error ellipsoid in (t_0, s_2) space?

c. Plot the error ellipsoid in the (t_0, s_2) plane and calculate conservative 95% confidence intervals for t_0 and s_2 for the appropriate value of Δ^2. Hint: The following MATLAB function will plot a two-dimensional covariance ellipse about the model parameters, where C is the covariance matrix, DELTA2 is Δ^2, and m is the 2-vector of model parameters.

```
%set the number of points on the ellipse to generate and plot
function plot_ellipse(DELTA2,C,m)
n=100;
%construct a vector of n equally-spaced angles from (0,2*pi)
```

```
theta=linspace(0,2*pi,n)';
%corresponding unit vector
xhat=[cos(theta),sin(theta)];
Cinv=inv(C);
%preallocate output array
r=zeros(n,2);
for i=1:n
%store each (x,y) pair on the confidence ellipse
%in the corresponding row of r
r(i,:)=sqrt(DELTA2/(xhat(i,:)*Cinv*xhat(i,:)'))*xhat(i,:);
end
plot(m(1)+r(:,1), m(2)+r(:,2));
axis equal
```

d. Evaluate the *p*-value for this model. You may find the library function **chi2cdf** to be useful here.

e. Evaluate the value of χ^2 for 1000 Monte Carlo simulations using the data prediction from your model perturbed by noise that is consistent with the data assumptions. Compare a histogram of these χ^2 values with the theoretical χ^2 distribution for the correct number of degrees of freedom. You may find the library function **chi2pdf** to be useful here.

f. Are your *p*-value and Monte Carlo χ^2 distribution consistent with the theoretical modeling and the data set? If not, explain what is wrong.

g. Use IRLS to find 1-norm estimates for t_0 and s_2. Plot the data predictions from your model relative to the true data and compare with (a).

h. Use Monte Carlo error propagation and IRLS to estimate symmetric 95% confidence intervals on the 1-norm solution for t_0 and s_2.

i. Examining the contributions from each of the data points to the 1-norm misfit measure, can you make a case that any of the data points are statistical outliers?

2. In this chapter we have largely assumed that the data errors are independent. Suppose instead that the data errors have an MVN distribution with expected value $\mathbf{0}$ and a covariance matrix $\mathbf{C}_D$. It can be shown that the likelihood function is then

$$L(\mathbf{m}|\mathbf{d}) = \frac{1}{(2\pi)^{m/2}} \frac{1}{\sqrt{\det(\mathbf{C}_D)}} e^{-(\mathbf{Gm}-\mathbf{d})^T \mathbf{C}_D^{-1} (\mathbf{Gm}-\mathbf{d})/2}. \tag{2.110}$$

a. Show that the maximum likelihood estimate can be obtained by solving the minimization problem,

$$\min (\mathbf{Gm} - \mathbf{d})^T \mathbf{C}_D^{-1} (\mathbf{Gm} - \mathbf{d}). \tag{2.111}$$

b. Show that (2.111) can be solved using the system of equations

$$\mathbf{G}^T \mathbf{C}_D^{-1} \mathbf{Gm} = \mathbf{G}^T \mathbf{C}_D^{-1} \mathbf{d}. \tag{2.112}$$

c. Show that (2.111) is equivalent to the linear least squares problem

$$\min \|\mathbf{C}_D^{-1/2}\mathbf{G}\mathbf{m} - \mathbf{C}_D^{-1/2}\mathbf{d}\|_2, \tag{2.113}$$

where $\mathbf{C}_D^{-1/2}$ is the matrix square root of $\mathbf{C}_D^{-1}$.

d. The Cholesky factorization of $\mathbf{C}_D^{-1}$ can also be used instead of the matrix square root. Show that (2.111) is equivalent to the linear least squares problem

$$\min \|\mathbf{R}\mathbf{G}\mathbf{m} - \mathbf{R}\mathbf{d}\|_2, \tag{2.114}$$

where $\mathbf{R}$ is the Cholesky factor of $\mathbf{C}_D^{-1}$.

3. Use MATLAB to generate 10,000 realizations of a data set of $m = 5$ points $\mathbf{d} = a + b\mathbf{x} + \boldsymbol{\eta}$, where $\mathbf{x} = [1, 2, 3, 4, 5]^T$, the $n = 2$ true model parameters are $a = b = 1$, and $\boldsymbol{\eta}$ is an m-element vector of independent $N(0, 1)$ noise.

a. Assuming that the noise standard deviation is known *a priori* to be 1, solve for the parameters a and b using least squares for each realization and histogram them in 100 bins.

b. Calculate the parameter covariance matrix, $\mathbf{C} = \sigma^2(\mathbf{G}^T\mathbf{G})^{-1}$, assuming independent $N(0, 1)$ data errors, and give standard deviations, σ_a and σ_b, for your estimates of a and b estimated from $\mathbf{C}$.

c. Calculate standardized parameter estimates

$$a' = \frac{a-1}{\sqrt{C_{1,1}}} \tag{2.115}$$

and

$$b' = \frac{b-1}{\sqrt{C_{2,2}}} \tag{2.116}$$

for your solutions for a and b. Demonstrate using a **Q–Q** plot (Appendix B) that your estimates for a' and b' are distributed as $N(0, 1)$.

d. Show, using a **Q–Q** plot, that the squared residual lengths

$$\|\mathbf{r}\|_2^2 = \|\mathbf{d} - \mathbf{G}\mathbf{m}\|_2^2 \tag{2.117}$$

for your solutions in (a) are distributed as χ^2 with $m - n = \nu = 3$ degrees of freedom.

e. Assume that the noise standard deviation for the synthetic data set is not known, and instead estimate it for each realization, k, as

$$s_k = \sqrt{\frac{1}{n-m}\sum_{i=1}^{m} r_i^2}. \tag{2.118}$$

Histogram your standardized solutions

$$a' = \frac{a - \bar{a}}{\sqrt{C'_{1,1}}} \tag{2.119}$$

and

$$b' = \frac{b - \bar{b}}{\sqrt{C'_{2,2}}}, \tag{2.120}$$

where $\mathbf{C}' = s_k^2(\mathbf{G}^T\mathbf{G})^{-1}$ is the covariance matrix estimation for the kth realization.

f. Demonstrate using a **Q–Q** plot that your estimates for a' and b' are distributed as the Student's t distribution with $\nu = 3$ degrees of freedom.

4. Suppose that we analyze a large number of data sets $\mathbf{d}$ in a linear regression problem and compute p-values for each data set. The χ^2_{obs} values should be distributed according to a χ^2 distribution with $m - n$ degrees of freedom. Show that the corresponding p-values will be uniformly distributed between 0 and 1.

5. Use linear regression to fit a polynomial of the form

$$y_i = a_0 + a_1 x_i + a_2 x_i^2 + \ldots + a_{19} x_i^{19} \tag{2.121}$$

to the noise-free data points

$$(x_i,\ y_i) = (-0.95,\ -0.95),\ (-0.85,\ -0.85),\ \ldots,\ (0.95,\ 0.95). \tag{2.122}$$

Use the normal equations to solve the least squares problem.

Plot the data and your fitted model, and list the parameters, a_i, obtained in your regression. Clearly, the correct solution has $a_1 = 1$, and all other $a_i = 0$. Explain why your answer differs.

2.8. NOTES AND FURTHER READING

Linear regression is a major subfield within statistics, and there are literally hundreds of associated textbooks. Many of these references focus on applications of linear regression in the social sciences. In such applications, the primary focus is often on determining which variables have an effect on response variables of interest (rather than on estimating parameter values for a predetermined model). In this context it is important to compare alternative regression models and to test the hypothesis that a predictor variable has a nonzero coefficient in the regression model. Since we normally know which predictor variables are important in the physical sciences, the approach commonly differs. Useful linear regression references from the standpoint of estimating parameters in the context considered here include [40, 113].

Robust statistical methods are an important topic. Huber discusses a variety of robust statistical procedures [77]. The computational problem of computing a 1-norm solution has been extensively researched. Techniques for 1-norm minimization include methods based on the simplex method for linear programming, interior point methods, and iteratively reweighted least squares [6, 32, 130, 138]. The IRLS method is the simplest to implement, but interior point methods can be the most efficient approaches for very large problems. Watson reviews the history of methods for finding p-norm solutions including the 1-norm case [171].

We have assumed that $\mathbf{G}$ is known exactly. In some cases elements of this matrix might be subject to measurement error. This problem has been studied as the **total least squares problem** [78]. An alternative approach to least squares problems with uncertainties in $\mathbf{G}$ that has recently received considerable attention is called **robust least squares** [11, 43].